

IMPROVEMENTS IN THE SYNTHESIS AND CHARACTERIZATION OF GLYCIDYL METHACRYLATE-GRAFTED EPDM THROUGH MELT FREE-RADICAL PROCESS

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RUBBER CHEMISTRY AND TECHNOLOGY, Vol. 00, No. 0, pp. 000–000 (0000)

ABSTRACT

Melt free-radical grafting reactions between EPDM and glycidyl methacrylate (GMA) were investigated in the presence or absence of styrene (St) comonomer in a batch mixer (170 °C, 60 rpm). The effects of dicumyl peroxide (DCP) initiator, GMA, and St concentrations were studied on the grafted EPDM characteristics. Torque–time profiles detected the grafting process by a melt viscosity rise. Titration results indicated an increase in the graft degree (GD) and gel content (GC) values with increasing of the DCP concentration. Introduction of St led to a 132% increase in GD and a 39% decrease in GC by preventing macroradicals from crosslinking side-reaction. Glass-transition temperatures of all samples increased compared with the pure EPDM due to the increase in GC. The increase was greater in the samples containing comonomer because of the stronger intermolecular interactions. Fourier-transform infrared spectroscopy spectra and contact-angle measurements confirmed that GMA was grafted onto EPDM. A calibration curve was developed as a criterion to predict the GD in melt free-radical process. [doi:10.5254/rct.19.80447]

INTRODUCTION

The inert nature of most polymers is the key characteristic allowing for their considerable use in a variety of applications, but a disadvantage for applications in which surface wettability or interaction with polar components is required.¹ To obviate this problem, several studies have been conducted on the functionalization of polymers (e.g., polyolefins, polyesters) through free-radical grafting of functional monomers onto the polymer backbone. Monomers mostly used for this purpose are maleic anhydride (MAH),^{2–8} itaconic anhydride,^{7,9,10} and glycidyl methacrylate (GMA).^{11–19} Free-radical reactions are initiated by primary free-radical generators such as 2,5-dimethyl-2,5-bis(*t*-butylperoxy)hexane,^{13–16} 2,5-dimethyl-2,5-di(*tert*iobutylperoxy)hexane,^{6,10} dicumyl peroxide (DCP),^{8,11,12,20–22} and benzoyl peroxide.^{2,18}

As a compatibilizer in polymer blends and composites,^{23,24} modified polyolefins containing epoxy function and succinic anhydride ring (for GMA and MAH, respectively) have the potential tendency to react with many functional groups such as hydroxyl, amine, and carboxyl.^{13,25} However, the reactivity of GMA-grafted polyolefins is reported to be higher compared with those modified with MAH.²⁵ A comparison of the role of MAH and GMA in the grafting efficiency of polyolefins shows that the reactivity of double bonds in MAH is lower due to the presence of neighbor electron-attracting carbonyl groups and therefore GMA reacts with free-radicals more easily.¹⁷

Free-radical grafting of functional monomers onto polyolefins is always accompanied with side reactions depending on the nature of macroradicals and the polyolefin. Chain branching and crosslinking and chain scission are the most common side reactions detected for polyethylene and polypropylene, respectively.^{4,9} Introducing a second monomer such as St,^{12,17,25–28} divinyl benzene (DVB),¹⁵ or trimethylolpropanetriacrylate (Tris)¹³ to the grafting reactions as “comonomer” or “co-agent” is suggested to control undesirable side reactions. Comonomer is

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shown to decrease the gel content (GC) dramatically where a small effect on the graft degree (GD) has been reported.³

There are several studies that have investigated the parameters effecting the free-radical functionalization reaction of high-density polyethylene,^{5,17,18} low-density polyethylene,^{3,25} polypropylene,^{29–31} linear low-density polyethylene,³² and EPM,^{12,13,15} whereas free-radical grafting onto EPDM has rarely been investigated. Malaika and Kong¹³ and Doudin et al.¹⁵ studied the effect of peroxide initiator, GMA concentration, and two different comonomer (i.e., Tris and DVB) on the grafting degree and gel content of functionalized EPM. It was observed that the grafting efficiency improved at the presence of Tris¹³ and DVB¹⁵ comonomer, requiring much lower peroxide initiator concentration. In fact, GMA homopolymerization, as the major competing reaction, was drastically reduced, leading to no detectable gel or crosslinking. Hu and Cartier¹² investigated the effect of St as a comonomer on the melt free-radical grafting of GMA onto EPM. They found that introduction of St to the graft reaction increased the grafting yield as well as the grafting rate. The effect of grafted GMA on the thermal properties of EPM was monitored by differential scanning calorimetry (DSC), revealing that crystallization temperature and melting enthalpy increased by grafting reaction. This was attributed to the role of grafted GMA as the crystallization nucleating agent, although melting temperature did not show any significant change.²² In the only paper published so far on the free-radical grafting process of EPDM, formation of poly-GMA and EPDM crosslinking have been detected as the undesirable side reactions. It was found that although Tris comonomer eliminated the formation of GMA homopolymers and gave rise to higher grafting yield produced at a much lower peroxide concentration, EPDM crosslinking reactions remained high. This was attributed to the multifunctionality and high reactivity of Tris toward both EPDM and GMA. Hence, Tris assisted long- and multibranched grafts of GMA onto EPDM chains rather than the preferred short grafted functions, thus reducing the efficiency of the graft functionalities in locating at the interface.¹⁴

According to the reviewed gaps in the grafting process of EPDM, St-assisted free-radical grafting reactions at different dicumylperoxide initiator and GMA concentrations were investigated in the present study. Grafting degree and gel content of EPDM samples functionalized at the absence and presence of St comonomer were analyzed through Fourier-transform infrared spectroscopy (FTIR) and titration techniques. A GMA–graft calibration curve is also developed for EPDM used here. Thermal characterization and contact angle measurements of were conducted using DSC and drop shape analyzer, respectively, to validate the previous results.

EXPERIMENTAL

MATERIALS

Materials used in this research were as follows: EPDM, KEP 270 (supplied by Kumho Polychem, Korea), with Mooney viscosity $ML_{(1+4)}$ @ 125 °C; 71M (ethylene content, 57%; Ter monomer content, 4.5% ethyldenenorbornene; density, 0.86 g/cm³); GMA monomer (97%, Aldrich Chemical Co., Japan); DCP (98%, Merck Millipore); and St monomer (99%, Merck Millipore). Xylene, acetone, ethanol, trichloroacetic acid (TCA), phenolphthalein, and potassium hydroxide used for titration and purification were all of analytical grade obtained from Merck Millipore.

PREPARATION

Melt grafting reactions were conducted using a 60 cc internal mixer (Brabender W50, Duisburg, Germany) preheated to 170 °C. The torque value was recorded for identically weighed

TABLE I
MELT FREE-RADICAL GRAFTING COMPONENTS: CONCENTRATIONS AND
NOMENCLATURE

Sample code	DCP, phr	GMA, phr	St, phr
S-St-1	0.5	3	2.25
S-St-2	0.5	5	3.75
S-St-3	0.5	7	5.25
Blank 1	0	5	0
S1	0.5	5	0
S2	0.3	5	0
Blank 2	0	3	0
S3	0.3	3	0
S4	0.15	3	0

samples as a function of mixing time. Regarding the volatility of GMA at the processing temperature, 20 min before the mixing, GMA, DCP initiator, and St chain-transfer agent (for St-assisted batches) were physically poured on EPDM in a closed container. Impregnated EPDM was then melt mixed for 8 min at 60 rpm. All EPDM-g-GMA samples were prepared according details in Table I. St/GMA molar ratio in S-St-1, S-St-2, and S-St-3 was set to 1.^{17,19,26}

CHARACTERIZATION

GC and GD Measurements. — A few grams of each EPDM-g-GMA sample was put in 100 mL of boiling xylene for 1 h. The remaining solid filtrate was dried in vacuum oven at 80 °C for 24 h. The GC was then calculated according to Eq. 1:

$$GC = \frac{w_1}{w_0} \times 100,$$

where w_0 and w_1 are sample weights before and after xylene solution, respectively.

The filtrate was stirred with acetone anti-solvent for 20 min when the solution became cloudy. To eliminate unreacted monomers, the cloudy solution was rinsed with acetone after filtration. The remnant is, in fact, purified EPDM-g-GMA.

Pure EPDM-g-GMA weighed sample was solved in boiling xylene, and TCA (0.3 M) was added to the solution before putting it in 90 °C oven, where the reaction between GMA epoxy rings and TCA would progress to completion. Reacted EPDM-g-GMA then settled, introducing acetone/ethanol (1:1). The filtrate was titrated using 0.1 M KOH and a few drops of phenolphthalein as the indicator. GD was calculated according to Eq. 2:

$$GD = \frac{142.15 \times (C_0 V_0 - C_2 V_2)}{W} \times 100,$$

where C_0 and C_2 is the TCA and KOH standard solution concentration, respectively; V_0 and V_2 is the TCA standard solution volume and KOH standard solution volume used through titration, respectively; and W indicates the initial sample weight.

An FTIR spectroscope (Bomem-102, Canada) was used to track the expected reactions. Samples were purified with boiling xylene and acetone/ethanol antisolvent before acetone rinsing. S-St-1, S-St-2, and S-St-3 were compression molded (FCHWABENPHAN400, Germany) at 220 °C to form a thin film (<100 µm) where S1, S2, S3, and S4 were solved in boiling xylene to prepare

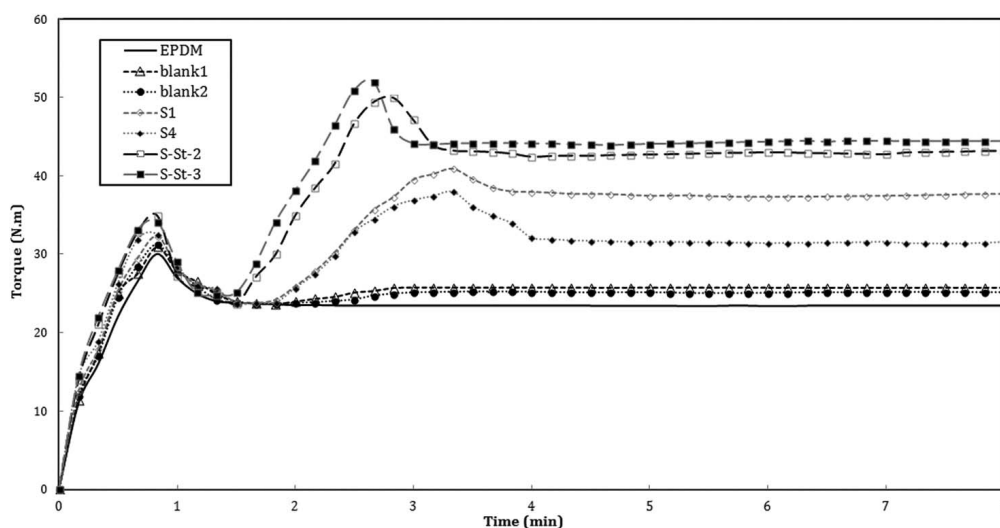


FIG. 1. — Torque vs mixing time for typical samples.

the test sample.^{7,32} The FTIR transmittance spectra were recorded with molded films and KBR pellets, respectively.

DSC was conducted using a Mettler Toledo DSC instrument (Stare software, Spain) under a nitrogen atmosphere. The temperature range of -100 to 25 °C with a heating rate of 10 °C/min was scanned. One of the most important transitions recognized by DSC technique is the glass transition, where the free volume increases and as a result the first segmental mobility in the amorphous phase of an amorphous or semi-crystalline polymer and increasing in the specific heat capacity (c_p) can be observed.^{33–35} DSC can be very useful to evaluate the glass-transition temperature (T_g) when T_g is well separated, sharp, and sufficiently large.³⁶

To study the wettability of samples, static contact angle measurements were conducted at 20 °C using a contact angle instrument running drop shape analysis DSA100 software (Krüss GmbH, Hamburg, Germany), equipped with a steel micro-syringe needle. A drop of the test liquid (water) was dispensed on the sample surface. Contact angle was recorded while monitoring the drop with a camera for 30 s.

RESULTS AND DISCUSSION

TORQUE RHEOMETRY

Typical mixing torque versus processing time profiles are given in Figure 1, demonstrating how melt viscosity revolves during the EPDM grafting reactions with different compositions. Pure EPDM experiences an initial rise for material loading before the torque levels off. Blank 1 and Blank 2 show a slight increase before the plateau, attributed to the partial crosslinking reactions increasing the mixture viscosity.³⁷ This is in accordance with the GC values measured for the blank samples presented in the following section. Two distinct peaks are distinguished for the reactive blends (i.e., S1, S4, S-St-2, and S-St-3), respectively corresponding to material loading and grafting reactions. In fact, grafting of GMA onto EPDM chains leads to stronger interactions between EPDM chains. A similar trend has been reported for grafted EPM.¹² Undescending plateau registered for all samples declines the polymer degradation hypothesis suggested by other publications^{12,38} within the mixing time studied here.

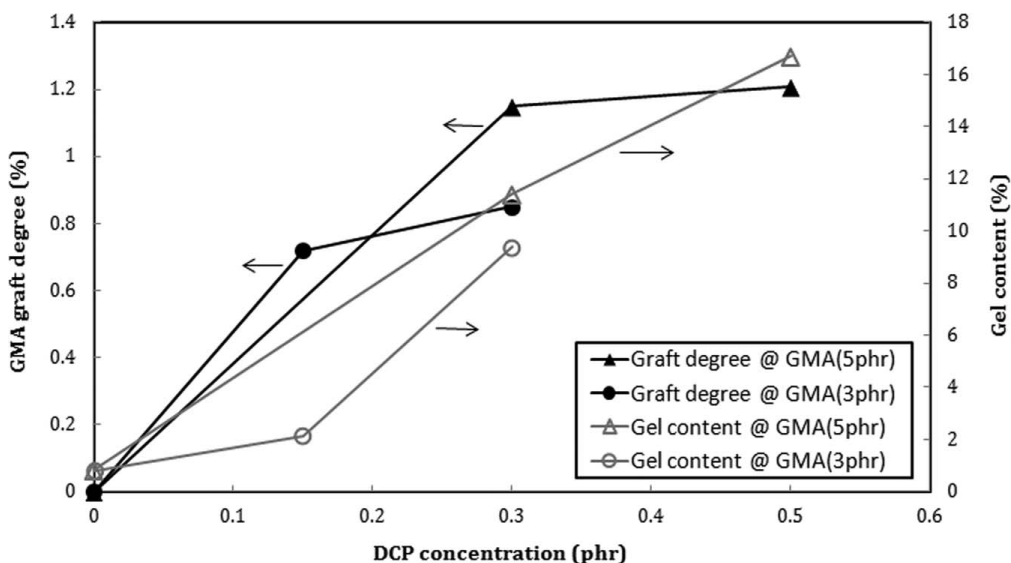


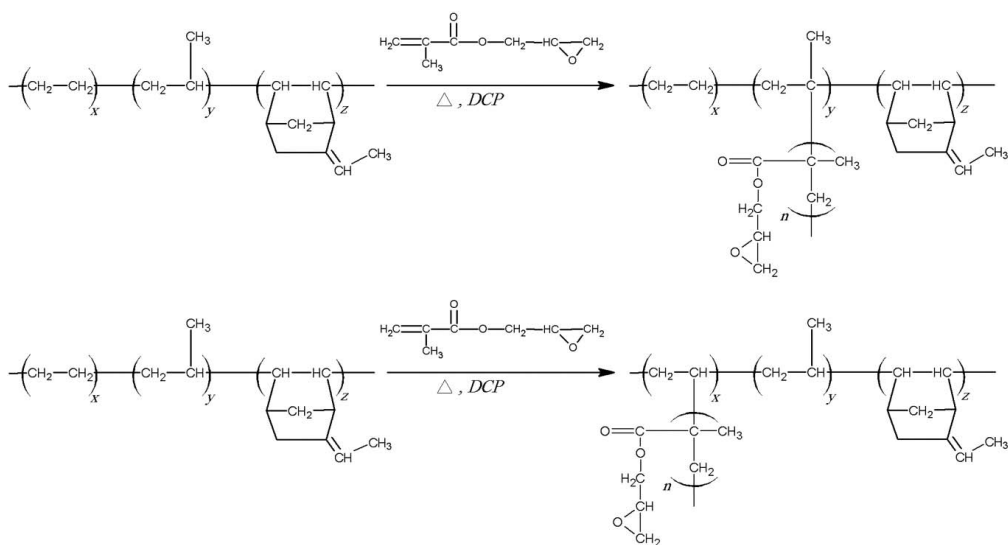
FIG. 2. — GMA grafted EPDM, GD, and GC vs DCP concentration at different GMA concentrations.

Comparing the intensity of grafting peak appeared in the reactive specimens showed higher viscosity levels for S-St-2 and S-St-3 prepared in the presence of St propounding higher grafted degrees compared with S1 and S4. As a result, the grafting reaction in the presence of St seems to start sooner, continue faster, and take less time to accomplish. As expected, the amount of torque drop to plateau has decreased at higher gel contents. These correlations are fully addressed in the next section.

GEL CONTACT AND DEGREE OF GRAFTING

Figure 2 compares the variations observed in GC and GMA GD with increasing DCP amount at fixed GMA contents, for those samples prepared with no St. At both GMA contents studied here (i.e., 3 and 5 parts per hundred of rubber [phr]); a similarly increasing trend is observed for GD values versus DCP content. This is attributed to the accelerated formation of free radicals and consequently macroradicals along the polymer chain at the presence of DCP. In fact, the DCP initiator undergoes a homolytic decomposition upon thermomechanical heating to yield a pair of primary free radicals³² that can abstract a tertiary hydrogen from the EPDM chain to form a macromolecular radical (Scheme 1).³⁹ These macroradicals could initiate the grafting of polymer chains or undergo undesirable competing reactions depending on the polymer backbone nature.¹² However, crosslinking might be more likely to happen as a competing pathway at the presence of higher free-radical counts, and this would attenuate the mentioned increasing trend as clearly observed in Figure 2.

GC has also increased with DCP content at the same time with the GD, but the trend is somehow different. At GMA content of 5 phr, the increasing rate is almost uniform along all DCP contents studied herein, whereas for S3 and S4 containing 3 phr GMA, a greater slope is observed at higher DCP contents when DCP amount changes from 0.15 to 0.3 phr because crosslinking competing reactions would be more contingent at the presence of higher concentrations of macroradicals. This would produce more gel-like structures in the sample bulk. At GMA contents of 5 phr, the increasing rate was observed to be unaffected by DCP concentration, which might be due to the homopolymerization of GMA molecules beside the crosslinking reactions. In such circumstances, a portion of the free radicals



SCHEME 1. — Schematic of graft reactions mechanism.

present in the reaction media are involved in the GMA homopolymerization. This would suppress the expected crosslinking reactions. Limited solubility of monomer in the polymer, yielding high local concentration, has also known to be responsible for oligo(GMA) grafts.⁴⁰

It can be concluded that increasing initiator concentration beneficially affects the improvements observed in GC values where the GD rate is slightly suppressed. Similar behavior has been reported by Hu and Cartier¹² studying the effect of DCP on the reactions grafting GMA onto EPM.

Comparing the values presented in Figure 2 and Table II for S-St-1, S-St-2, and S-St-3 with those of S1, S2, S3, S4, Blank 1 and Blank 2 shows that the GD and GC of the samples prepared at the presence of St chain-transfer agent are greater and smaller, respectively, than the samples prepared without St. It should be noted that the direct attack of the macroradicals to GMA is significantly hindered by the steric effect of polymer chains, whereas in the presence of St comonomer, the EPDM macroradicals preferably attack St due to its less steric effect. Produced styryl radicals then are copolymerized with GMA monomers easily. By contrast, St could progress a reaction with EPDM macroradicals much faster than GMA, and this would prevent the macroradicals to be involved in the crosslinking competing pathways. The resulting styryl macroradicals are also stabilized by resonance of St. This procedure would cause an increase in the GD.^{12,27,32} In this way, GMA is not directly grafted onto EPDM chains and St acts as an intermediate. Considering the mentioned role for St, GC is expected to decrease at the presence of St comonomer, as observed here. It is believed that when St is added, more GMA is converted to grafted species onto EPDM chains; thus, the amount of GMA

TABLE II
GD AND GC OF EPDM SAMPLES GRAFTED AT PRESENCE OF ST (DCP = 0.5
PHR)

Sample code	GD, %	Graft efficiency, %	GC, %
S-St-1	1.93	64.3	8.51
S-St-2	2.81	56.2	10.12
S-St-3	3.04	43.4	10.68

available for homopolymerization would reduce.⁴¹ Al-Malaika and Kong,^{13,14} contradictorily observed that the presence of trimethylolpropanetriacrylate chain-transfer agent does increase the GMA GD on EPDM, but GC unexpectedly remains high. This behavior was attributed to the crosslinking of this multifunctional comonomer in the reaction media. However, in a similar condition, GC of GMA-grafted EPM samples was reported to dramatically decrease in the presence of this comonomer.^{13,15} Brito et al.³² believed that GMA grafting reaction of LLDPE at the presence of GMA, St, and redox initiator (peroxide [DCP]/reducing agent [Sn(Oct)₂]) yields to the greatest extent of GDs (maximal GD $\approx$ 4.7% @ 47% graft efficiency) by preventing cage effect, although they did not address the other key factor, GC.

The mean graft conversion (i.e., grafted GMA monomers to the primary GMA content) is as low as 25%, which is ascribed to the unfavorable electrostatic and esteric hindrance effect in the grafting reactions.³² It is also interesting to note that graft conversion is greater at lower GMA concentrations. This confirms the higher probability of GMA homopolymerization side reactions at higher GMA concentrations, evidenced in the above-mentioned discussions. Graft efficiency has increased with DCP concentration, as expected.

Table II illustrates the variations in GC and GD of St-assisted graft samples versus GMA and St concentrations. GD is observed to increase with simultaneous increment of GMA and St concentration where its increasing rate has been diminished at higher GMA and St concentrations. Increasing GMA concentration, solely, can promote the homopolymerizations reaction of GMA monomers besides the graft reaction. By contrast, macroradicals are suppressed from entering the side reactions including crosslinking and GMA homopolymerization by St molecules. Thus, it is expected that increasing the St concentration from 3 to 7 phr will individually direct the monomers/macroradicals to participate in the main graft reaction. It can be concluded that at St/GMA molar ratio of unity suggested by similar research,^{17,19,26} increasing GMA and St content, within the range studied here, would be beneficial to GD, but not to graft efficiency, raising the homopolymerization side reactions between GMA monomers.

According to Table II, a moderate increase can be seen in the GC by simultaneously increasing equimolar amounts of GMA and St, and also the graft efficiency has decreased. Although GMA (positive on GD, negative on GC) and St (positive on GD and GC) effects have been individually investigated, the equimolar amount of GMA introduced to the reaction seems to negatively affect the reaction. Thus, although the molar concentration of St has been suggested to be equal to the concentration of graft monomer by most similar publications,^{12,25,26,32,42} the evidence detected in this work does not confirm this notion. Accordingly, to achieve acceptable GCs and graft efficiencies, St/GMA molar ratios greater than 1 seem to be more sensible. Considering that St, besides the direct reaction with macroradicals, could activate the graft monomer (i.e., GMA) by forming a charge-transfer complex,³² introducing St/GMA molar ratios less than 1 might restrict the St comonomer roles. Flat⁴¹ correspondingly showed that grafting yield does not always increase with St content, but it goes through a maximum when St/GMA molar ratio is between 1 and 2.

FTIR AND CALIBRATION CURVE

Figure 3 compares the FTIR results of pure EPDM and the EPDM samples grafted by GMA in the absence of St comonomer. As expected, the main characteristic peak confirming the formation of C=O carbonyl groups during the grafting reaction has appeared at 1730 cm⁻¹ for samples S1, S2, S3, and S4. Although Bray et al.⁴³ believed that formation of oligo(maleic anhydride) grafts would shift the characteristic peak of simple alkyl succinates, no evidence reliable enough to claim the formation of graft length was detected here. The epoxy group, responsible for the compatibilizing features of GMA-grafted polymers, was detected as a shoulder at 990 cm⁻¹.

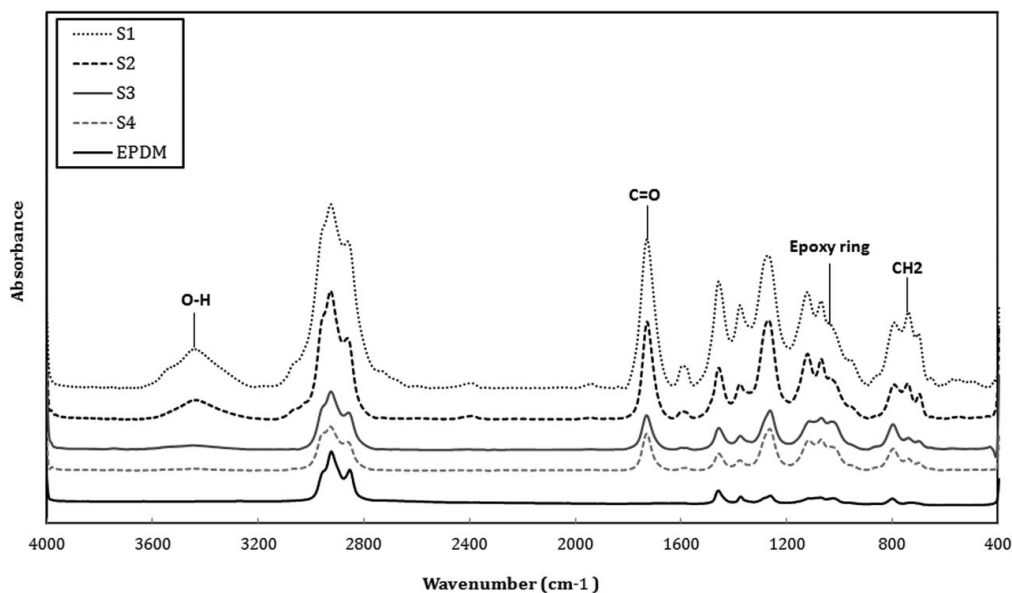


FIG. 3. — FTIR results of pure EPDM and the EPDM-g-GMA samples in the absence of St.

FTIR results of EPDM-g-GMA samples prepared in the presence of St comonomer are illustrated in Figure 4. Peaks appeared at 721, 972, 1734, and 3433, respectively, corresponding to stretching CH₂, epoxide, carbonyl, and hydroxyl functions, confirming the functionalization of EPDM chains by GMA.

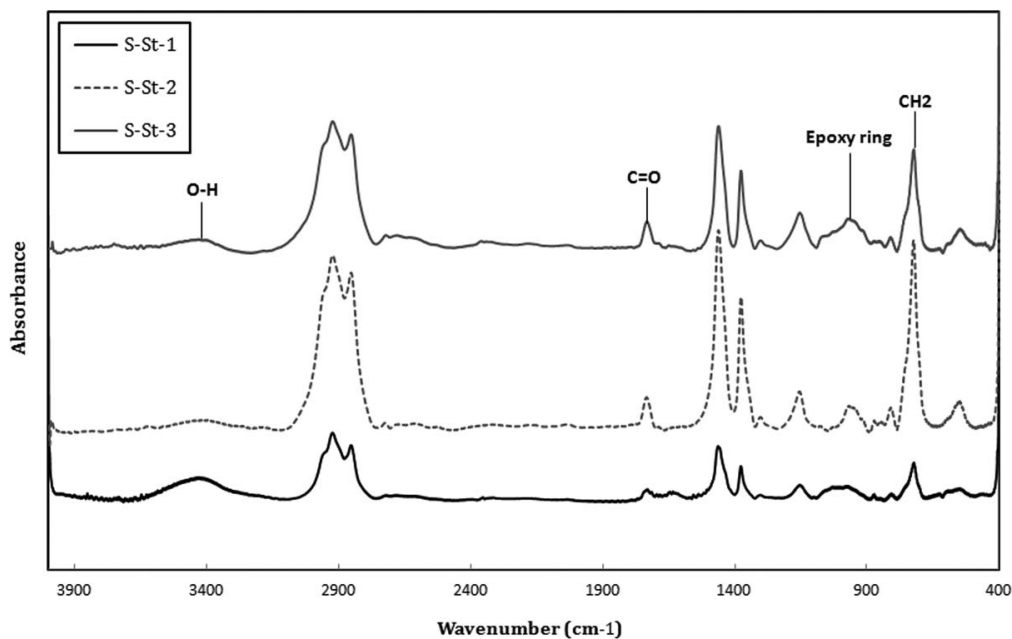


FIG. 4. — FTIR results of pure EPDM and the EPDM-g-GMA samples in the presence of St.

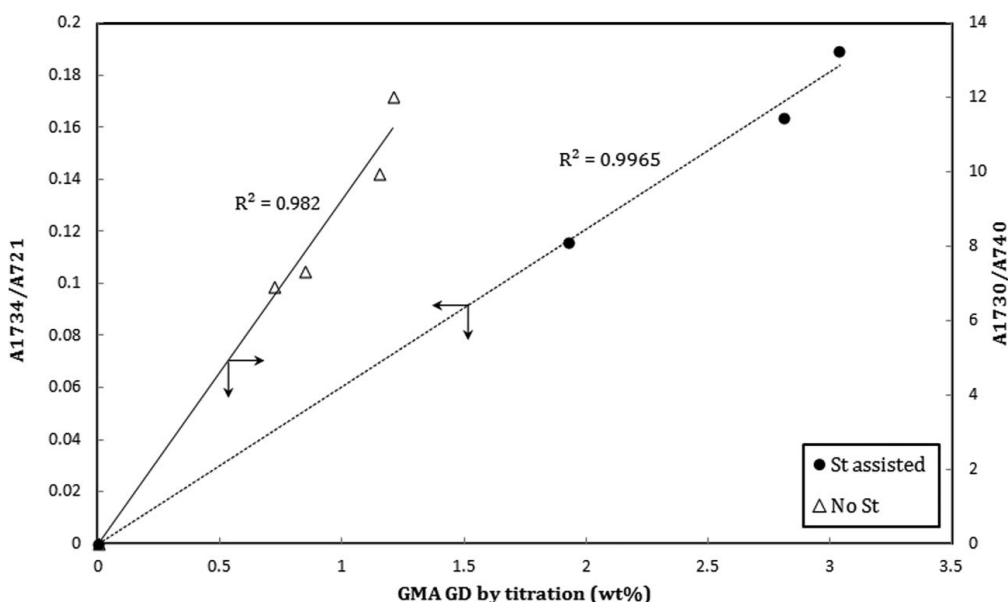


FIG. 5. — Titration-FTIR graft degree calibration curve for EPDM-g-MA prepared with (●) and without (Δ) St comonomer.

A calibration curve (Figure 5) was developed to obtain GDs on the basis of FTIR results. Similar curves were published by Papke and Karger-Kocsis,²⁰ Brito et al.,³² and Wei et al.^{13,25} This curve could provide an easy direct way to predict the GD that would be achieved through melt free-radical process. For this purpose, the carbonyl to stretching CH₂ peak area (A_{1730}/A_{740} and A_{1734}/A_{721}) was prescribed as the GD criterion. The absolute error calculated on the data ($R^2 = 0.981$ and 0.996) is acceptable. St-assisted process shows less deviation from the calibration line compared with the reaction conducted with no St. This could be ascribed to the unfavorable competing reactions suppressed at the presence of St. The equations below are obtained for melt free-radical grafting of GMA onto EPDM from the calibration curves:

For the St-assisted reaction,

$$GD = 16.556 \times \frac{A_{1734}}{A_{721}}$$

In the absence of St,

$$GD = 0.108 \times \frac{A_{1730}}{A_{740}}$$

PHYSICAL APPEARANCE

The physical appearance of EPDM-g-GMA samples prepared at different conditions was compared. Samples S1, S2, and S3 grafted without St were colored (light yellow), whereas sample coloration was reduced to white when grafting was conducted at the presence of St comonomer. Similar observations has been reported in the literature, attributing this phenomenon to unreacted DCPs, poly(St-co-GMA), and poly-GMA produced through the side reactions.^{41,44} Among the samples prepared without comonomer, S4 is exceptionally white, which could be due to the low

TABLE III
COMPARISON OF GLASS TRANSITION TEMPERATURES AND SPECIFIC HEAT
CAPACITY VALUES

Sample code	T_g , °C	Δc_p , J/g, °C
EPDM	−51.5	0.25
S1	−47.06	0.32
S2	−47.2	0.34
S3	−46.53	0.21
S4	−46.65	0.19
S-St-1	−40.33	0.18
S-St-2	−41.77	0.18
S-St-3	−47.61	0.31

measured GC. It was also interesting to observe that samples with low GDs were translucent compared with the opaque appearance of highly grafted samples.

DSC RESULTS

Thermal studies on the prepared samples were carried out using DSC. The glass transition is well recognizable in the form of a downhill in the second heating cycles (data not shown) from −150 °C to room temperature. Glass-transition temperatures and the specific heat capacity (Δc_p) values for all specimens including pure EPDM and differently grafted EPDM rubbers are summarized in Table III.

For the samples prepared in the absence (S1–S4) and presence (S-St-1 and S-St-2) of St comonomer, the average T_g is −46.86 and −41.05 °C, respectively. These values are 4.64 and 10.45 °C, respectively, greater than the equivalent transition for pure EPDM. It is well known that any structure or phenomenon that limits the chains mobility will increase the T_g value.³³ In this case, the increase observed for all samples in T_g compared with that of pure EPDM could be attributed to the GC growth as well as the inter/intramolecular forces established by grafting GMA on EPDM backbone.^{45–49} In fact, chain mobility restrictions induced by crosslinking and/or the mentioned interactions are responsible for this observation. Consequent effects are also seen for Δc_p values.

Table III shows that the mean T_g value for both S-St-1 and S-St-2 samples is approximately 6 °C beyond that of S1–S4 samples. The mean GC of S-St-1 and S-St-2 (see Table II) is not that different from S1 to S4 samples (9.32 and 9.91 %, respectively), but the GD of the samples prepared in the presence of comonomer is around twice the other modified rubbers. Therefore, it can be concluded that the influence of interactions dominants and leads to less segmental mobility freedom that consequently increases the T_g value.^{48,49} It is interesting that the T_g value for S-St-3 (−47.61 °C) is so close to the T_g value of modified rubbers prepared without comonomer. This could be ascribed to the large amount of unreacted St and GMA monomers decreasing the T_g value.

A weak endothermic peak beyond the glass-transition region is also observed for samples S1–S4 and S-St-4. It was believed that this peak can be attributed to the chain disentanglements, which need sufficiently large free volume to advance at a measurable rate.³³ This peak has also been attributed to EPDM melting overlap with the T_g region.⁵⁰

CONTACT ANGLES

Contact angles of water droplet against the surfaces of pure EPDM and EPDM-g-GMA samples are illustrated versus GDs in Figure 6. There is a monotonically decreasing trend in contact

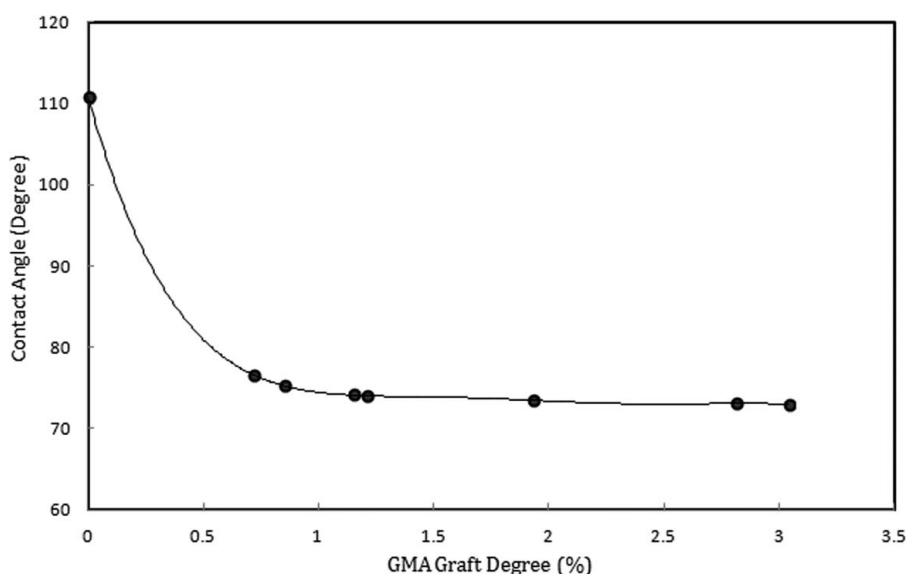


FIG. 6. — Contact angles vs graft degree for EPDM and EPDM-g-GMA samples.

angles with measured GDs, indicating that grafting GMA onto the EPDM surface has improved the wettability characteristics of the rubber, as expected. At higher GDs, the advancing rate is diminished, which could suggest that the grafting phenomenon has proceeded deeper down in the bulk.⁵¹

CONCLUSIONS

Glycidyl methacrylate was grafted onto ethylene–propylene–diene terpolymer through melt free-radical grafting process in the absence and presence of styrene chain-transfer agent. Graft degree and GC results obtained by titration indicated that styrene improves the graft degree by preventing the unfavorable crosslinking reactions. The decrease in GC values confirms this observation. Comparing the results published on the application of trimethylolpropanetriacrylate shows that styrene shows a better performance decreasing the GC besides its positive effect on the graft degree. In the absence of styrene, increasing the concentration of DCP can improve the graft degree as well as the GC. Analyzing the graft degree and GC results showed that sample S4 (DCP = 0.15 phr, GMA = 3 phr) with grafting degree of 0.72 wt% and GC of 2.12 wt% could be selected as the best sample to use as a compatibilizer in polymer blends. Thermal characterization (DSC), FTIR, and contact angle results all confirmed the conclusions based on titration technique. A calibration curve was developed on the basis of FTIR and titration results as a criterion to predict the graft degree through melt free-radical grafting reactions.

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[Received February 2019, Revised May 2019]